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Phase change random access memory device

Abstract

A method for making a memory device, includes: forming a first dielectric layer over a bottom electrode; forming a first void extending through the first dielectric layer to expose a portion of an upper boundary of the bottom electrode; forming a first conductive structure lining along respective sidewalls of the first void and the exposed portion of the upper boundary of the bottom electrode; filling the first void with the first dielectric layer; and forming a phase change material layer over the first dielectric layer to cause the phase change material layer to contact at least a portion of a sidewall of the first conductive structure.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 18/231,750, filed Aug. 8, 2023, which is a continuation of U.S. patent application Ser. No. 17/571,260, filed Jan. 7, 2022, which is a continuation of U.S. patent application Ser. No. 16/688,976, filed Nov. 19, 2019, now U.S. Pat. No. 11,233,197, which is a divisional application of U.S. patent application Ser. No. 15/904,064, filed Feb. 23, 2018, now U.S. Pat. No. 10,510,954, which claims priority to U.S. Provisional Patent Application No. 62/591,318, filed on Nov. 28, 2018, each of which are incorporated by reference herein in their entireties.

BACKGROUND

(1) In recent years, unconventional nonvolatile memory (NVM) devices, such as ferroelectric random access memory (FRAM) devices, resistive random access memory (RRAM) devices, and phase change random access memory (PCRAM) devices have emerged. In particular, PCRAM devices, which exhibit a switching behavior between a high resistance state and a low resistance state, have various advantages over conventional NVM devices. Such advantages include, for example, compatible fabrication steps with current complementary-metal-oxide-semiconductor (CMOS) technologies, low-cost fabrication, a compact structure, flexible scalability, fast switching, high integration density, etc.

(2) Generally, a PCRAM device includes a top electrode (e.g., an anode) and a bottom electrode (e.g., a cathode) with a phase change material layer interposed therebetween. Further, the bottom electrode is coupled to the phase change material layer with a conductive structure, typically known as a “heater” structure. To transition the PCRAM device to the low resistance state, which is typically referred to as a set operation, a relatively low electrical current signal is applied on the phase change material layer through the heater structure to anneal the phase change material layer at a temperature between respective crystallization (lower) and melting (higher) temperatures of the phase change material layer so as to crystallize the phase change material layer; and to transition the PCRAM device to the high resistance state, which is typically referred to as a reset operation, a relatively high electrical current signal is applied on the phase change material layer via the heater structure to anneal the phase change material layer at a temperature higher than the melting (higher) temperature of the phase change material layer so as to amorphize the phase change material layer. In particular, a current level of the applied electrical current signal that can successfully amorphize/crystallize the phase change material layer is proportional to a contact area size at an interface between the heater structure and the phase change material layer. For example, the bigger the contact area size is, the higher the current level of the applied electrical current signal needs to be.

(3) The heater structures of existing PCRAM devices, however, couple respective phase change material layers with relatively large contact areas, which disadvantageously requires respective current levels to be relatively high. Various issues may accordingly occur in existing PCRAM devices when applying such a high current level signal, for example, less reliability, higher power

consumption, etc. Thus, existing PCRAM devices and methods to make the same are not entirely satisfactory.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that various features are not necessarily drawn to scale. In fact, the dimensions and geometries of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. 1A and 1B illustrate a flow chart of an exemplary method for forming a semiconductor device, in accordance with some embodiments.

(3) FIG. 2A, 2B, 2C, 2D, 2E, 2F, 2G, 2H, 2I, 2J, 2K, 2L, 2M, 2N, and 2O illustrate cross-sectional views of an exemplary semiconductor device during various fabrication stages, made by the method of FIG. 1, in accordance with some embodiments.

(4) FIGS. 3A, 3B, 3C, 3D, 3E, and 3F illustrate respective plan views of various embodiments of part of the exemplary semiconductor device made by the method of FIG. 1, in accordance with some embodiments.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

(5) The following disclosure describes various exemplary embodiments for implementing different features of the subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(6) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(7) The present disclosure provides various embodiments of a novel PCRAM (phase change random access memory) device and methods to form the same. In some embodiments, the disclosed PCRAM device includes a bottom electrode, a bottom conductive column structure, a phase change material layer, an optional top conductive column structure, and a top electrode, wherein the bottom and top conductive column structures each includes a sidewall portion formed as a “shell” structure surrounding a “core” structure formed of a dielectric material. Specifically, the bottom electrode is coupled to the phase change material layer through the sidewall portion of the bottom conductive column structure, and the top electrode is coupled to the phase change material layer through the top conductive column structure. In some embodiments, the bottom and top conductive column structures may be each configured to conduct a current signal to the phase change material layer. Since the sidewall portion of the bottom conductive column structure is formed as the shell structure, a corresponding contact area size at the interface between the phase change material layer and the bottom conductive column structure can be significantly reduced

when compared to the conventional PCRAM devices. As such, various issues that the conventional PCRAM devices encounter can be advantageously avoided in disclosed PCRAM devices.

(8) FIGS. 1A and 1B illustrate a flowchart of a method **100** to form a semiconductor device according to one or more embodiments of the present disclosure. It is noted that the method **100** is merely an example, and is not intended to limit the present disclosure. In some embodiments, the semiconductor device is, at least part of, a PCRAM device. As employed by the present disclosure, the PCRAM device refers to any device including a phase change material layer. It is noted that the method **100** of FIGS. 1A and 1B does not produce a completed PCRAM device. A completed PCRAM device may be fabricated using complementary metal-oxide-semiconductor (CMOS) technology processing. Accordingly, it is understood that additional operations may be provided before, during, and after the method **100** of FIGS. 1A and 1B, and that some other operations may only be briefly described herein. In some other embodiments, the method may be used to form any of a variety of nonvolatile memory (NVM) devices, such as ferroelectric random access memory (FRAM) devices, resistive random access memory (RRAM) devices, etc., while remaining within the scope of the present disclosure.

(9) Referring first to FIG. 1A, in some embodiments, the method **100** starts with operation **102** in which a substrate including a transistor is provided. The method **100** continues to operation **104** in which a first dielectric layer including a contact plug extending therethrough is formed over the substrate. In some embodiments, the first dielectric layer is formed over the transistor, and the contact plug is electrically coupled to at least one conductive feature (e.g., a drain, source, or a gate feature) of the transistor. The method **100** continues to operation **106** in which a second dielectric layer is formed over the first dielectric layer. The method **100** continues to operation **108** in which a first electrode is formed in the second dielectric layer. In some embodiments, the first electrode is electrically coupled to the contact plug extending through the first dielectric layer. The method **100** continues to operation **110** in which a third dielectric layer is formed over the second dielectric layer.

(10) Next, the method **100** continues to operation **112** in which a portion of the third dielectric layer is etched to form a trench extending through the third dielectric layer so as to expose a portion of an upper boundary of the first electrode. The method **100** continues to operation **114** in which an isolation layer is formed over the etched third dielectric layer to line the trench. As such, the isolation layer extends along sidewalls of the trench and overlays a bottom boundary of the trench (i.e., the portion of the upper boundary of the first electrode that was exposed in operation **112**). It is noted that in some embodiments, the isolation layer may also overlay an upper boundary of the third dielectric layer. The method **100** continues to operation **116** in which a portion the isolation layer is etched to re-expose the portion of the upper boundary of the first electrode. In some embodiments, concurrently with the exposure of the portion of the upper boundary of the first electrode, another portion of the isolation that overlays the upper boundary of the third dielectric layer is also etched away.

(11) Referring then to FIG. 1B, the method **100** continues to operation **118** in which a conductive layer is formed over the etched third dielectric layer to line the trench. Specifically, in some embodiments, the conductive layer is formed to overlay the re-exposed portion of the upper boundary of the first electrode, extend along sidewalls of the trench (with the isolation layer coupled therebetween), and overlays the upper boundary of the third dielectric layer. The method **100** continues to operation **120** in which a dielectric material is formed over the etched third dielectric layer to fill the trench. Specifically, in some embodiments, the dielectric material fills the trench with the isolation layer and the conductive layer disposed therebetween. The method **100** continues to operation **122** in which a polishing process is performed to form a first conductive column structure. In some embodiments, the polishing process (e.g., a chemical mechanical polishing (CMP) process) is performed on the dielectric material and the conductive layer until an upper boundary of a sidewall portion of the conductive layer is exposed, which accordingly forms

the first conductive column structure. As such, the first conductive column structure includes at least two portions: a first portion that is the portion of the conductive layer overlaying the upper boundary of the first electrode; and a second portion that is the sidewall portion of the conductive layer.

(12) Next, the method **100** continues to operation **124** in which a phase change material layer is formed over the first conductive column structure. In some embodiments, the phase change material layer is formed to couple at least the upper boundary of the sidewall portion of the conductive layer (the first conductive column structure). The method **100** continues to operation **126** in which a fourth dielectric layer is formed over the phase change material layer. The method **100** continues to operation **128** in which a second conductive column structure is formed to couple the phase change material layer. In some embodiments, the second conductive column structure, which may be optionally formed in the fourth dielectric layer, is substantially similar to the first conductive column structure. The method **100** continues to operation **130** in which a second electrode is formed to couple the second conductive column structure. In some embodiments, the second electrode, which may be formed in a fifth dielectric layer over the fourth dielectric layer, is substantially similar to the first electrode. In some embodiments, the above-mentioned first, second, third, fourth, and fifth dielectric layers may each be an inter-metal dielectric (IMD) or inter-layer dielectric (ILD) layer, that is, the first, second, third, fourth, and fifth dielectric layers may be formed of a substantially similar dielectric material (e.g., low-k dielectric material).

(13) In some embodiments, operations of the method **100** may be associated with cross-sectional views of a semiconductor device **200** at various fabrication stages as shown in FIGS. 2A, 2B, 2C, 2D, 2E, 2F, 2G, 2H, 2I, 2J, 2K, 2L, 2M, 2N, and 2O, respectively. In some embodiments, the semiconductor device **200** may be a PCRAM device. The PCRAM device **200** may be included in a microprocessor, memory cell, and/or other integrated circuit (IC). Also, FIGS. 2A through 2O are simplified for a better understanding of the concepts of the present disclosure. For example, although the figures illustrate the PCRAM device **200**, it is understood the IC, in which the PCRAM device **200** is formed, may include a number of other devices comprising resistors, capacitors, inductors, fuses, etc., which are not shown in FIGS. 2A through 2O, for purposes of clarity of illustration.

(14) Corresponding to operation **102** of FIG. 1A, FIG. 2A is a cross-sectional view of the PCRAM device **200** including a substrate **202** with a transistor **204**, which is provided at one of the various stages of fabrication, according to some embodiments. Although the PCRAM device **200** in the illustrated embodiment of FIG. 2A includes only one transistor **204**, it is understood that the illustrated embodiment of FIG. 2A and the following figures are merely provided for illustration purposes. Thus, the PCRAM device **200** may include any desired number of transistors while remaining within the scope of the present disclosure.

(15) In some embodiments, the substrate **202** includes a semiconductor material substrate, for example, silicon. Alternatively, the substrate **202** may include other elementary semiconductor material such as, for example, germanium. The substrate **202** may also include a compound semiconductor such as silicon carbide, gallium arsenic, indium arsenide, and indium phosphide. The substrate **202** may include an alloy semiconductor such as silicon germanium, silicon germanium carbide, gallium arsenic phosphide, and gallium indium phosphide. In one embodiment, the substrate **202** includes an epitaxial layer. For example, the substrate may have an epitaxial layer overlying a bulk semiconductor. Furthermore, the substrate **202** may include a semiconductor-on-insulator (SOI) structure. For example, the substrate may include a buried oxide (BOX) layer formed by a process such as separation by implanted oxygen (SIMOX) or other suitable technique, such as wafer bonding and grinding.

(16) In some embodiments, the transistor **204** includes a gate electrode **204-1**, a gate dielectric layer **204-2**, and source/drain features **204-3** and **204-4**. The source/drain features **204-3** and **204-4** may be formed using doping processes such as ion implantation. The gate dielectric layer **204-2**

may include a dielectric material such as, silicon oxide, silicon nitride, silicon oxynitride, dielectric with a high dielectric constant (high-k), and/or combinations thereof, which may be formed using deposition processes such as atomic layer deposition (ALD). The gate electrode **204-1** may include a conductive material, such as polysilicon or a metal, which may be formed using deposition processes such as chemical vapor deposition (CVD). In some embodiments, the transistor **204** may serve as an access transistor of the PCRAM device **200**, which controls an access to a data storage component (e.g., a PCRAM resistor) of the PCRAM device **200** during read/write operations.

(17) Corresponding to operation **104** of FIG. 1A, FIG. 2B is a cross-sectional view of the PCRAM device **200** including a first dielectric layer **208** with a contact plug **210**, which is formed at one of the various stages of fabrication, according to some embodiments. As shown, the first dielectric layer **208** is formed over the transistor **204**, and the contact plug **210** is formed to extend through the first dielectric layer **206**. In some embodiments, the contact plug **210** is coupled to at least one of the conductive features of the transistor **204**. In the illustrated embodiment of FIG. 2B (and the following figures), the contact plug **210** is coupled to the source/drain feature **204-3**.

(18) In some embodiments, the first dielectric layer **208** is formed of a dielectric material. Such a dielectric material may include at least one of: silicon oxide, a low dielectric constant (low-k) material, other suitable dielectric material, or a combination thereof. The low-k material may include fluorinated silica glass (FSG), phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), carbon doped silicon oxide ($\text{SiO}_2\text{C}_x\text{O}_y$), strontium oxide (SrO), Black Diamond® (Applied Materials of Santa Clara, Calif.), Xerogel, Aerogel, amorphous fluorinated carbon, Parylene, BCB (bis-benzocyclobutenes), SiLK (Dow Chemical, Midland, Mich.), polyimide, and/or other future developed low-k dielectric materials. In some embodiments, the contact plug **210** is formed of a conductive material such as, for example, copper (Cu), aluminum (Al), tungsten (W), etc.

(19) The contact plug **210** may be formed by at least some of the following process steps: using chemical vapor deposition (CVD), physical vapor deposition (PVD), spin-on coating, and/or other suitable techniques to deposit the above-described dielectric material of the first dielectric layer **208** over the substrate **202** and the transistor **204**; performing one or more patterning processes (e.g., a lithography process, a dry/wet etching process, a cleaning process, a soft/hard baking process, etc.) to form an opening through the dielectric material; using CVD, PVD, E-gun, and/or other suitable techniques to deposit the above-described conductive material to refill the opening; and polishing out excessive conductive material to form the contact plug **210**.

(20) Corresponding to operation **106** of FIG. 1A, FIG. 2C is a cross-sectional view of the PCRAM device **200** including a second dielectric layer **212**, which is formed at one of the various stages of fabrication, according to some embodiments. As shown, the second dielectric layer **212** is formed over the first dielectric layer **208** and the contact plug **210**.

(21) In some embodiments, the second dielectric layer **212** is formed of a dielectric material. Such a dielectric material may include at least one of: silicon oxide, a low dielectric constant (low-k) material, other suitable dielectric material, or a combination thereof. The low-k material may include fluorinated silica glass (FSG), phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), carbon doped silicon oxide ($\text{SiO}_2\text{C}_x\text{O}_y$), strontium oxide (SrO), Black Diamond® (Applied Materials of Santa Clara, Calif.), Xerogel, Aerogel, amorphous fluorinated carbon, Parylene, BCB (bis-benzocyclobutenes), SiLK (Dow Chemical, Midland, Mich.), polyimide, and/or other future developed low-k dielectric materials. The second dielectric layer **212** may be formed by using chemical vapor deposition (CVD), physical vapor deposition (PVD), spin-on coating, and/or other suitable techniques to deposit the above-described dielectric material of the second dielectric layer **212** over the first dielectric layer **208**.

(22) Corresponding to operation **108** of FIG. 1A, FIG. 2D is a cross-sectional view of the PCRAM device **200** including a first electrode **214**, which is formed at one of the various stages of fabrication, according to some embodiments. As shown, the first electrode **214** is embedded in the

second dielectric layer **212** and horizontally extends along the second dielectric layer **212**. In some embodiments, the first electrode **214** is coupled to the contact plug **210**, and as will be discussed below, the first electrode **212** may serve as a bottom electrode of the data storage component (e.g., a PCRAM resistor) of the PCRAM device **200**.

(23) In some embodiments, the first electrode **214** is formed of a conductive material such as, for example, copper (Cu), aluminum (Al), tungsten (W), etc. The first electrode **214** may be formed by at least some of the following process steps: performing one or more patterning processes (e.g., a lithography process, a dry/wet etching process, a cleaning process, a soft/hard baking process, etc.) to form an opening through the second dielectric layer **212** so as to expose at least a portion of the contact plug **210**; using CVD, PVD, E-gun, and/or other suitable techniques to deposit the above-described conductive material to refill the opening; and polishing out excessive conductive material to form the first electrode **214**.

(24) Corresponding to operation **110** of FIG. **1A**, FIG. **2E** is a cross-sectional view of the PCRAM device **200** including a third dielectric layer **216**, which is formed at one of the various stages of fabrication, according to some embodiments. As shown, the third dielectric layer **216** is formed over the second dielectric layer **212** and the first electrode **214**.

(25) In some embodiments, the third dielectric layer **216** is formed of a dielectric material. Such a dielectric material may include at least one of: silicon oxide, a low dielectric constant (low-k) material, other suitable dielectric material, or a combination thereof. The low-k material may include fluorinated silica glass (FSG), phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), carbon doped silicon oxide (SiO₂x₂C₂y₂), strontium oxide (SrO), Black Diamond® (Applied Materials of Santa Clara, Calif.), Xerogel, Aerogel, amorphous fluorinated carbon, Parylene, BCB (bis-benzocyclobutenes), SiLK (Dow Chemical, Midland, Mich.), polyimide, and/or other future developed low-k dielectric materials. The third dielectric layer **216** may be formed by using chemical vapor deposition (CVD), physical vapor deposition (PVD), spin-on coating, and/or other suitable techniques to deposit the above-described dielectric material of the third dielectric layer **216** over the second dielectric layer **212**.

(26) Corresponding to operation **112** of FIG. **1A**, FIG. **2F** is a cross-sectional view of the PCRAM device **200** in which a portion of the third dielectric layer **216** is etched at one of the various stages of fabrication, according to some embodiments. As shown, after the portion of the third dielectric layer **216** is etched, a trench, or void, **218** that extends through the third dielectric layer **216** is formed. Accordingly, in some embodiments, the trench **218** exposes inner sidewalls **216'** of the third dielectric layer **216** and at least a portion of an upper boundary **214'** of the first electrode **214**.

(27) In some embodiments, the trench **218** is formed by performing at least some of the following process steps: performing one or more patterning processes (e.g., a lithography process, a dry/wet etching process, a cleaning process, a soft/hard baking process, etc.) to form a patternable layer with an opening over the third dielectric layer **216**, wherein the opening is laterally aligned with at least a portion of the upper boundary **214'** of the first electrode **214**; performing one or more dry/wet etching process on the third dielectric layer **216** while using the patternable layer as a mask until the portion of the upper boundary **214'** is exposed; and removing the patternable layer.

(28) Corresponding to operation **114** of FIG. **1A**, FIG. **2G** is a cross-sectional view of the PCRAM device **200** including an isolation layer **222**, which is formed at one of the various stages of fabrication, according to some embodiments. As shown, the isolation layer **222** is formed to overlay an upper boundary **216''** of the third dielectric layer **216**, and line the trench **218** (i.e., extending along the sidewalls **216'** and overlaying the upper boundary **214'**). In some embodiments, the isolation layer **222** is substantially thin and conformal so that the profile of the trench **218** may remain present after the formation of the isolation layer **222**.

(29) In some embodiments, the isolation layer **222** is formed of a dielectric material, for example, silicon oxide, silicon nitride, or the like. The isolation layer **222** may be formed by using chemical vapor deposition (CVD), physical vapor deposition (PVD), spin-on coating, and/or other suitable

techniques to deposit the above-described dielectric material of the isolation layer **222** over the etched third dielectric layer **216**.

(30) Corresponding to operation **116** of FIG. **1A**, FIG. **2H** is a cross-sectional view of the PCRAM device **200** in which an etching process **223** is performed at one of the various stages of fabrication, according to some embodiments. According to some embodiments, the etching process **223** may be an anisotropic etching process (e.g., a reactive ion etching (RIE) process). Thus, concurrently with or subsequently to the etching process **223**, portions of the isolation layer **222** that overlay the upper boundaries **214'** of the first electrode **214** and **216''** of the third dielectric layer **216**, respectively, are removed, which leaves sidewall portions **222-1** of the isolation layer **222** intact. As shown in the illustrated embodiment of FIG. **2H** (and the following figures), the sidewall portions **222-1** extends along the sidewall **216'** of the third dielectric layer **216**.

(31) Corresponding to operation **118** of FIG. **1B**, FIG. **2I** is a cross-sectional view of the PCRAM device **200** including a conductive layer **224**, which is formed at one of the various stages of fabrication, according to some embodiments. As shown, the conductive layer **224** is formed to line the trench **218** and overlay the upper boundary **216''**. More specifically, the conductive layer **224** lines the trench **218** by extending along the sidewall portions **222-1** of the isolation layer **222** with respective sidewall portions **224-1** and overlaying the upper boundary **214'** of the first electrode **214** with a bottom portion **224-2**. In some embodiments, the conductive layer **224** is substantially thin and conformal (e.g., 1 nanometers to 10 nanometers) so that the profile of the trench **218** may remain present after the formation of the conductive layer **224**.

(32) Although in FIG. **2I** (and the following figures), the conductive layer **224** is illustrated as a single layer, it is understood that the conductive layer **224** may include two or more layers stacked on the top of one another, each of which may be formed of a conductive material such as, for example, gold (Au), platinum (Pt), ruthenium (Ru), iridium (Ir), titanium (Ti), aluminum (Al), copper (Cu), tantalum (Ta), tungsten (W), iridium-tantalum alloy (Ir—Ta), indium-tin oxide (ITO), or any alloy, oxide, nitride, fluoride, carbide, boride or silicide of these, such as TaN, TiN, TiAlN, TiW, or a combination thereof. In some embodiments, the first conductive layer **224** is formed by using chemical vapor deposition (CVD), plasma enhanced (PE) CVD, high-density plasma (HDP) CVD, inductively-coupled-plasma (ICP) CVD, physical vapor deposition (PVD), spin-on coating, and/or other suitable techniques to deposit the at least one of the above-described conductive material over the etched third dielectric layer **216** and the sidewall portions **222-1** of the isolation layer **222**.

(33) Corresponding to operation **120** of FIG. **1B**, FIG. **2J** is a cross-sectional view of the PCRAM device **200** including a dielectric material **228**, which is deposited at one of the various stages of fabrication, according to some embodiments. As shown, the dielectric material **228** is formed to overlay the conductive layer **224**, which accordingly fills the trench **218** with the dielectric material **228**.

(34) In some embodiments, the dielectric material **228** is formed of a substantially similar dielectric material as the third dielectric layer **216**. Such a dielectric material may include at least one of: silicon oxide, a low dielectric constant (low-k) material, other suitable dielectric material, or a combination thereof. The low-k material may include fluorinated silica glass (FSG), phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), carbon doped silicon oxide (SiO.sub.xC.sub.y), strontium oxide (SrO), Black Diamond® (Applied Materials of Santa Clara, Calif.), Xerogel, Aerogel, amorphous fluorinated carbon, Parylene, BCB (bis-benzocyclobutenes), SiLK (Dow Chemical, Midland, Mich.), polyimide, and/or other future developed low-k dielectric materials. The dielectric material **228** may be formed by using chemical vapor deposition (CVD), physical vapor deposition (PVD), spin-on coating, and/or other suitable techniques to deposit the above-described dielectric material over the conductive layer **224**.

(35) Corresponding to operation **122** of FIG. **1B**, FIG. **2K** is a cross-sectional view of the PCRAM device **200** including a first conductive column structure **230**, which is formed at one of the various

stages of fabrication, according to some embodiments. According to some embodiments, the first conductive column structure **230** is formed by performing at least one CMP process on the dielectric material **228** and the conductive layer **224** disposed thereunder (FIG. **2J**) until respective upper boundaries **225** of the sidewall portions **222-1** are exposed. Accordingly, a remaining portion **228'** of the dielectric material **228** is surrounded by the sidewall portions **224-1** and bottom portion **224-2** of the conductive layer **224** at its sidewalls and a bottom boundary, which exposes a respective upper boundary **229**. In other words, the first conductive column structure **230** may include a first portion **224-1** formed as a shell structure (hereinafter “shell portion **224-1**”) surrounding the remaining portion **228'** formed as a core structure, and a second portion **224-2** coupled to one end of such a shell structure (hereinafter “end portion **224-2**”).

(36) Corresponding to operation **124** of FIG. **1B**, FIG. **2L** is a cross-sectional view of the PCRAM device **200** including a phase change material layer **232**, which is formed at one of the various stages of fabrication, according to some embodiments. As shown, the phase change material layer **232** is formed over the etched third dielectric layer **216** to couple at least part of the first conductive column structure **230**. In the illustrated embodiment of FIG. **2L**, the phase change material layer **232** is coupled to both the shell portions **224-1** of the first conductive column structure **230**. More specifically, the phase change material layer **232** overlays both the boundaries **225** of the shell portions **224-1**. In some other embodiments, the phase change material layer **232** may be only coupled to part of the shell portions **224-1** (e.g., one of the shell portions **224-1**), which will be illustrated and discussed below with respect to FIGS. **3A-3F**.

(37) In some embodiments, the phase change material layer **232** includes a chalcogenide-based material. Chalcogens include any of the four elements oxygen (O), sulfur (S), selenium (Se), and tellurium (Te), forming part of group VI of the periodic table. Chalcogenides comprise compounds of a chalcogen with a more electropositive element or radical. Chalcogenide alloys comprise combinations of chalcogenides with other materials such as transition metals. A chalcogenide alloy usually contains one or more elements from column six of the periodic table of elements, such as germanium (Ge) and tin (Sn). For example, chalcogenide alloys include combinations including one or more of antimony (Sb), gallium (Ga), indium (In), and silver (Ag). Exemplary materials of the phase change material layer **232** include alloys of: Ga/Sb, In/Sb, In/Se, Sb/Te, Ge/Te, Ge/Sb/Te, In/Sb/Te, Ga/Se/Te, Sn/Sb/Te, In/Sb/Ge, Ag/In/Sb/Te, Ge/Sn/Sb/Te, Ge/Sb/Se/Te and Te/Ge/Sb/S.

(38) In some embodiments, the phase change material layer **232** may be formed by an atomic layer deposition (ALD) technique with a precursor containing a metal and oxygen. In some embodiments, other chemical vapor deposition (CVD) techniques may be used. In some embodiments, the phase change material layer **232** may be formed by a physical vapor deposition (PVD) technique, such as a sputtering process with a metallic target and with a gas supply of oxygen and optionally nitrogen to the PVD chamber. In some embodiments, the phase change material layer **232** may be formed by an electron-beam deposition technique.

(39) Corresponding to operation **126** of FIG. **1B**, FIG. **2M** is a cross-sectional view of the PCRAM device **200** including a fourth dielectric layer **234**, which is formed at one of the various stages of fabrication, according to some embodiments. As shown, the fourth dielectric layer **234** is formed over the etched third dielectric layer **216** to overlay the phase change material layer **232**.

(40) In some embodiments, the fourth dielectric layer **234** is formed of a dielectric material. Such a dielectric material may include at least one of: silicon oxide, a low dielectric constant (low-k) material, other suitable dielectric material, or a combination thereof. The low-k material may include fluorinated silica glass (FSG), phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), carbon doped silicon oxide (SiO₂.xSi₃N₄.y), strontium oxide (SrO), Black Diamond® (Applied Materials of Santa Clara, Calif.), Xerogel, Aerogel, amorphous fluorinated carbon, Parylene, BCB (bis-benzocyclobutenes), SiLK (Dow Chemical, Midland, Mich.), polyimide, and/or other future developed low-k dielectric materials. The fourth dielectric layer **234** may be formed by using chemical vapor deposition (CVD), physical vapor deposition (PVD), spin-on

coating, and/or other suitable techniques to deposit the above-described dielectric material of the fourth dielectric layer **234** over the etched third dielectric layer **216** and the phase change material layer **232**.

(41) Corresponding to operation **128** of FIG. **1B**, FIG. **2N** is a cross-sectional view of the PCRAM device **200** including a second conductive column structure **240**, which is formed at one of the various stages of fabrication, according to some embodiments. As shown, the second conductive structure **240** extends through part of the fourth dielectric layer **234** to couple the phase change material layer **232**.

(42) In some embodiments, the second conductive column structure **240** is substantially similar to the first conductive column structure **230** so that the configuration of the second conductive column structure **240** is briefly discussed below, and the formation of the second conductive column structure **240** is not repeated here. The second conductive column structure **240** also includes a shell portion **242-1** surrounding a core structure formed of a remaining portion **241** of a dielectric material (similar as the dielectric material **228** of FIG. **2J**), and an end portion **242-2** coupled to one end of the shell portion **242-1**. As such, the second conductive column structure **240** is coupled to the phase change material layer **232** via the end portion **242-2**, and to the fourth dielectric layer **234** via the sidewall portion **242-1** with an isolation layer **242-1** disposed therebetween. In some embodiments, the isolation layer **242-1** is substantially similar to the sidewall portion **222-1**. Further, although in the illustrated embodiment of FIG. **2N** the second conductive column structure **240** is aligned with the first conductive column structure **230**, it is noted that the second conductive column structure **240** may be laterally shifted from the first conductive column structure **230** (as long as the second conductive column structure **240** is still coupled to the phase change material layer **232**) while remaining within the scope of the present disclosure.

(43) Corresponding to operation **130** of FIG. **1B**, FIG. **2O** is a cross-sectional view of the PCRAM device **200** including a second electrode **244**, which is formed at one of the various stages of fabrication, according to some embodiments. As shown, the second electrode **244** is embedded in a fifth dielectric layer **246** and horizontally extends along the fifth dielectric layer **246**. In some embodiments, the second electrode **244** is coupled to the second conductive column structure **240**, and as will be discussed below, the second electrode **212** may serve as a top electrode of the data storage component (e.g., a PCRAM resistor) of the PCRAM device **200**.

(44) In some embodiments, the second electrode **244** is formed of a conductive material such as, for example, copper (Cu), aluminum (Al), tungsten (W), etc. The second electrode **244** may be formed by at least some of the following process steps: performing one or more patterning processes (e.g., a lithography process, a dry/wet etching process, a cleaning process, a soft/hard baking process, etc.) to form an opening through the fifth dielectric layer **246** so as to expose at least a portion of the second conductive column structure **240**; using CVD, PVD, E-gun, and/or other suitable techniques to deposit the above-described conductive material to refill the opening; and polishing out excessive conductive material to form the second electrode **244**. It is noted that in some embodiments, each of the above-described first/second/third/fourth/fifth dielectric layers (**208/212/216/234/246**) may be an inter-metal dielectric (IMD) or inter-layer dielectric (ILD) layer.

(45) In some embodiments, after the formation of the second electrode **244**, a PCRAM resistor of the PCRAM device **200** may be formed. More specifically, the first electrode **214** may serve as the bottom electrode of the PCRAM resistor, the first conductive column structure **230** may serve as the heater structure of the PCRAM resistor, the phase change material layer **232** may be configured to switch between the low and high resistance states by being transitioned to the partially crystalline and amorphous states, respectively, the second conductive column structure **240** may serve as an optional heater structure of the PCRAM resistor, and the second electrode **244** may serve as the top electrode.

(46) In operation, the PCRAM device **200** may be “granted” to be accessed through the access

transistor **204**. Upon accessed, the PCRAM device **200** may transition between the low and high resistance states through set and reset operations, respectively, as discussed above. When compared to the conventional PCRAM devices, a contact area size between the heater structure (e.g., the first conductive column structure **230**) and the phase change material layer **232** is significantly reduced. Specifically, in the illustrated embodiment of FIG. 20, the contact area size may be defined as twice the cross-sectional area of the shell portion **224-1** (i.e., the thickness of the conductive layer **224**, as described with respect to FIG. 21), which is significantly smaller than the contact area size of the conventional PCRAM devices that typically includes additional cross-sectional area of the remaining portion **228'**. Accordingly, the current level of an electrical current signal applied to transition the resistance states of the phase change material layer **232** can be advantageously reduced, which avoids various issues that the conventional PCRAM device are facing.

(47) FIGS. 3A, 3B, 3C, 3D, 3E, and 3F respectively illustrate top views of various embodiments of how the first conductive column structure **230** that extends through the dielectric layer **216** is spatially configured with respect to the phase change material layer **232** that is disposed in another dielectric layer above the dielectric layer **216**. As will be discussed below, the shell portion **224-1** of the first conductive column structure **230** may be formed as a circular or polygonal ring when viewed from the top; the phase change material layer **232** may be formed to have circular shape, a polygonal shape, or a circumferential shape when viewed from the top; and the phase change material layer **232** may be partially overlapped with (i.e., coupled to) a portion of the shell portion **224-1** of the first conductive column structure **230**.

(48) For example, in FIG. 3A, the shell portion **224-1** is formed as a rectangular ring that surrounds the remaining portion **228'** of the dielectric material **228**. Alternatively stated, the shell portion **224-1** has an inner boundary and an outer boundary that each form a circumference of a rectangle shape. And the phase change material layer **232**, formed as a rectangular shape, overlaps (e.g., contacts) part of such a rectangular ring. Accordingly, a corresponding contact area size may be defined as a cross-sectional area of overlapping **301** enclosed by dotted lines. In FIG. 3B, the shell portion **224-1** is formed as a rectangular ring that is substantially similar to the one shown in FIG. 3A, but the phase change material layer **232**, also formed as a rectangular shape, overlaps the shell portion **224-1** by a smaller area (a cross-sectional area of overlapping **303** enclosed by dotted lines).

(49) In FIG. 3C, the shell portion **224-1** is formed as a circular ring surrounds the remaining portion **228'** of the dielectric material **228**. Alternatively stated, the shell portion **224-1** has an inner boundary and an outer boundary that each form a circumference of a circular shape. And the phase change material layer **232**, formed as a rectangular ring (e.g., a circumferential shape), overlaps part of such a circular ring. Accordingly, a corresponding contact area size may be defined as a cross-sectional area of overlapping **305** enclosed by dotted lines.

(50) In FIG. 3D, the shell portion **224-1** is formed as a first circular ring surrounds the remaining portion **228'** of the dielectric material **228**, and the phase change material layer **232** (enclosed by dotted lines) is formed as a second circular ring. Further, in some embodiments, such a first circular ring (the shell portion **224-1**) may overlap with the second circular ring (the phase change material layer **232**) by running along a circumference of the second circular ring, which defines an overlapping (not shown) that is part of the cross-sectional area of either the first or second circular ring.

(51) In FIG. 3E, the shell portion **224-1** is formed as a rectangular ring and the phase change material layer **232** is formed as a rectangular shape, which is similar to FIGS. 3A and 3B, except that the phase change material layer **232** may be formed to have two or more respective different portions. For example, in the illustrated embodiment of FIG. 3E, the phase change material layer **232** includes a first portion **232-1** and a second portion **232-2** that are laterally spaced apart from each other and respectively overlapped with the rectangular ring of the shell portion **224-1**.

(52) In FIG. 3F, the shell portion **224-1** is formed as a circular ring and the phase change material

layer 232 is formed to have two rectangle shapes that are laterally spaced apart from each other. For example, in the illustrated embodiment of FIG. 3F, the phase change material layer 232 includes a first portion 232-3 and a second portion 232-4 that are respectively overlapped with the circular ring of the shell portion 224-1.

(53) In an embodiment, a memory device includes: a first conductive column structure extending through a first dielectric layer, wherein the first conductive column structure comprises a shell portion wrapping a core structure filled with a dielectric material and an end portion that is coupled to one end of the shell portion and disposed below the core structure; and a first phase change material layer formed over the first dielectric layer, wherein a lower boundary of the first phase change material layer contacts at least a first portion of the other end of the shell portion of the first conductive column structure.

(54) In another embodiment, a memory device includes: a bottom electrode; a phase change material layer; and a heater structure, coupled between the bottom electrode and the phase change material layer, that lines a trench thereby causing a first portion of the heater structure to contact the phase change material layer at one end and wrap a dielectric material directly disposed below the phase change material layer.

(55) Yet in another embodiment, a method includes: forming a first dielectric layer over a bottom electrode; forming a first void extending through the first dielectric layer to expose a portion of an upper boundary of the bottom electrode; forming a first conductive structure lining along respective sidewalls of the first void and the exposed portion of the upper boundary of the bottom electrode; filling the first void with the first dielectric layer; and forming a phase change material layer over the first dielectric layer to cause the phase change material layer to contact at least a portion of a sidewall of the first conductive structure.

(56) The foregoing outlines features of several embodiments so that those ordinary skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A memory device, comprising: a first conductive column structure extending through a first dielectric layer; a first phase change material layer formed over the first dielectric layer; a second dielectric layer disposed above the first phase change material layer; and a second conductive column structure extending through the second dielectric layer, wherein the second conductive column structure comprises a shell portion wrapping a core structure filled with a dielectric material and an end portion that is coupled to a first end of the shell portion.
2. The memory device of claim 1, further comprising a bottom electrode disposed below the first dielectric layer, and coupled to the end portion of the first conductive column.
3. The memory device of claim 2, further comprising a top electrode disposed above the second dielectric layer, and coupled to a second end of the shell portion of the second conductive column, the second end being opposite the first end.
4. The memory device of claim 1, wherein the first conductive column structure also comprises a shell portion, and wherein the shell portion of the first conductive column structure has a cross-sectional area of about 1 to 10 nanometers.
5. The memory device of claim 1, wherein the first conductive column structure also comprises a shell portion, and wherein the shell portion of the first conductive column structure comprises an

inner boundary and an outer boundary that each form a circumference of a circular shape when viewed from top.

6. The memory device of claim 5, wherein the shell portion of the first conductive column structure comprises an inner boundary and an outer boundary that each form a circumference of a polygonal shape when viewed from top.

7. The memory device of claim 5, further comprising: a second phase change material layer formed over the first dielectric layer, wherein a lower boundary of the second phase change material layer contacts at least a second portion of the other end of the shell portion of the first conductive column.

8. The memory device of claim 7, wherein at least one of the first and second phase change material layers has a circular shape, a polygonal shape, or a circumferential shape when viewed from top.

9. A memory device, comprising: a bottom electrode formed in a first dielectric layer; a phase change material layer formed in a second dielectric layer; and a heater structure formed in a third dielectric layer and extending at least partially through the third dielectric layer between the bottom electrode and the phase change material layer, wherein a first portion of the heater structure contacts the phase change material layer at a first end and wraps a dielectric material disposed below the phase change material layer.

10. The memory device of claim 9, wherein the first portion of the heater structure comprises an inner boundary and an outer boundary that each form a circumference of a circular shape when viewed from top.

11. The memory device of claim 9, wherein the heater structure is formed of a conductive material selected from at least one of: TaN, TiN, TiAlN, and TiW.

12. The memory device of claim 9, wherein the first portion of the heater structure has a cross-sectional area of about 1 to 10 nanometers.

13. The memory device of claim 9, further comprising: a top electrode disposed above and coupled to the phase change material layer.

14. The memory device of claim 9, wherein the first portion of the heater structure comprises an inner boundary and an outer boundary that each form a circumference of a polygonal shape when viewed from top.

15. The memory device of claim 9, wherein the phase change material layer has a circular shape, a polygonal shape, or a circumferential shape when viewed from top.

16. A memory device, comprising: a first conductive column structure extending through a first dielectric layer, wherein the first conductive column structure comprises a first shell portion wrapping a first core structure filled with a first dielectric material and a first end portion that is coupled to a first end of the first shell portion; a first phase change material layer formed over the first dielectric layer; a second dielectric layer disposed above the first phase change material layer; and a second conductive column structure extending through the second dielectric layer, wherein the second conductive column structure comprises a second shell portion wrapping a second core structure filled with a second dielectric material and a second end portion that is coupled to one end of the second shell portion and disposed below the second core structure.

17. The memory device of claim 16, further comprising a bottom electrode disposed below the first dielectric layer, and coupled to the first end portion of the first conductive column.

18. The memory device of claim 16, further comprising a top electrode disposed above the second dielectric layer, and coupled to the other end of the second shell portion of the second conductive column.

19. The memory device of claim 16, wherein the first end portion is disposed below the first core structure, and wherein the first shell portion comprises a conductive layer.

20. The memory device of claim 16, wherein an upper boundary of the first phase change material layer contacts the second end portion of the second conductive column.

